# Scale, proportions, circles, and review

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. $12 \times 3$

2. $12 \times 7$

3. $12 \times 4$

4. $12 \times 9$

5. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

6. 4 markers cost $2.60. Find dollars per marker.

7. Find $10 - 1.286$.

8. A prism has right-triangle ends with sides 3, 4, 5 cm and length 6 cm. Find surface area.

9. Order $-4$, 0, $-9$ from least to greatest.

10. Find the MAD of 1, 3, 5, 7.

11. A 8 by 12 cm rectangle is copied with scale factor $\frac{1}{2}$. Find the new side lengths.

12. A map uses 1 cm for 20 km. A route is 4 cm long. Find the actual distance.

13. A shape has area 20 square cm. All lengths scale by $\frac{1}{2}$. Find the new area.

14. A proportional table contains (6, 9). Find k in $y = kx$.

15. Eli triples every length of a drawing. He says its area also triples. Explain and repair his claim.

16. A rectangle 5 cm by 4 cm is scaled by 2. Find new dimensions and area.

17. A drawing has area 20 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

18. A map uses 1 cm for 5 km. A route is 4.5 cm on the map. Find actual distance.

## CHALLENGE / Explain your plan and check the result

C1. A model bridge is 48 cm long at a scale of 1 cm for 2.5 m. Find the real length. A new model uses 1 cm for 4 m. Find the new model length and the scale factor from the first model to the second.

C2. A recipe uses 3 cups of flour for 8 muffins. A baker needs 20 muffins and has 6 cups. Find the required flour and the shortage. If flour costs $0.80 per cup, find the cost of exactly the missing amount.